

NONG KHAI, ThailandWMO: **483520**Lat: **17.8650N**Lon: **102.7472E**Elev: **176**StdP: **99.23**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	15.6	7.8	6.7	19.3	9.6	7.6	19.4	5.0	22.2	4.3	23.7	0.6	90	0.296

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.2	38.0	25.5	36.7	25.6	35.5	25.6	27.8	33.1	27.5	32.6	27.2	32.1	1.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.7	29.9	26.2	22.1	29.5	26.1	21.9	29.3	90.3	33.5	88.6	32.8	87.3	32.3	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.3	3.6	3.2	DB	11.2	39.8	1.8	1.2	9.9	40.7	8.9	41.4	7.9	42.1	6.5	43.0	
			WB	9.5	28.7	1.7	0.7	8.3	29.3	7.3	29.7	6.3	30.1	5.1	30.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	23.3	25.5	28.2	30.1	29.5	29.1	28.5	28.2	27.7	25.9	23.4
		DBStd	3.08	2.55	3.04	2.83	2.46	1.98	1.56	1.56	1.37	1.35	1.60	2.70
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0
		HDD18.3	5	2	1	1	0	0	0	0	0	0	0	2
		CDD10.0	6315	412	435	564	603	606	572	574	563	546	549	476
		CDD18.3	3278	156	202	306	353	348	322	316	304	296	291	226
		CDH23.3	35614	1239	2010	3578	4566	4184	3779	3478	3238	3184	3051	2054
		CDH26.7	15343	406	908	1841	2516	2007	1630	1336	1153	1178	1196	782
(14)	Wind	WSAvg	0.8	0.9	0.9	0.9	0.8	0.8	0.7	0.8	0.8	0.8	0.9	1.0
(15)	Precipitation	PrecAvg	1711	10	18	46	89	233	276	317	340	282	94	6
		PrecMax	2237	73	79	130	246	462	507	536	657	508	242	41
		PrecMin	1216	0	0	1	26	65	96	184	201	129	3	0
		PrecStd	243	18	19	37	56	89	85	92	102	93	61	27
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	36.1	38.5	40.2	38.8	36.6	35.8	34.5	34.3	34.4	33.0
			MCWB	22.5	23.1	24.3	25.7	26.7	27.0	27.2	27.0	26.7	26.2	23.8
		2%	DB	31.6	34.6	37.2	38.6	36.9	35.0	34.1	33.4	33.4	32.9	31.5
			MCWB	22.1	22.9	24.2	25.8	26.5	27.0	27.0	26.8	26.4	25.5	23.1
		5%	DB	30.4	33.4	35.8	37.2	35.5	33.8	33.0	32.5	32.6	32.5	30.2
			MCWB	21.7	22.9	24.5	25.9	26.5	26.9	26.8	26.6	26.2	25.1	22.2
		10%	DB	29.1	32.0	34.4	35.7	34.0	32.8	32.0	31.5	31.7	31.7	29.0
			MCWB	21.1	22.8	24.3	25.7	26.3	26.7	26.6	26.5	26.0	24.7	21.5
		0.4%	WB	24.7	25.6	27.0	27.7	28.0	28.2	28.0	27.7	27.2	26.1	25.2
			MCDB	30.2	32.7	34.8	35.6	34.9	33.4	33.2	32.3	32.3	32.5	30.7
		2%	WB	23.5	24.7	25.9	27.1	27.4	27.7	27.6	27.4	27.2	26.6	24.0
			MCDB	29.0	31.8	33.4	35.0	33.8	32.7	32.1	31.5	31.4	31.2	29.3
(31)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	22.7	24.0	25.4	26.6	27.1	27.3	27.2	27.1	26.9	26.1	24.8
			MCDB	28.2	30.8	32.8	34.3	33.1	32.1	31.4	31.0	30.8	30.3	28.6
		10%	WB	21.9	23.4	24.8	26.2	26.7	27.0	26.8	26.7	26.6	25.7	24.2
			MCDB	27.4	29.6	32.0	33.5	32.3	31.5	30.7	30.3	30.2	29.6	27.7
		5% DB	MDBR	11.2	11.6	11.1	10.2	8.2	7.0	6.4	6.2	6.5	7.7	9.4
			MCDBR	12.1	13.0	13.0	12.1	10.3	8.5	8.0	7.8	7.9	8.4	10.7
(36)	Mean Daily Temperature Range	5% WB	MCWBR	4.1	3.5	3.4	3.0	2.8	2.5	2.5	2.6	2.4	2.3	3.5
			MCDBR	10.7	11.1	11.4	10.8	9.0	7.7	7.1	7.0	7.1	7.4	9.8
		5% WB	MCWBR	4.1	3.5	3.5	3.3	3.0	2.8	2.6	2.7	2.6	2.4	3.0
			MCDBR	10.7	11.1	11.4	10.8	9.0	7.7	7.1	7.0	7.1	7.4	9.8
		5% WB	MCWBR	4.1	3.5	3.5	3.3	3.0	2.8	2.6	2.7	2.6	2.4	3.0
(40)	Clear-Sky Solar Irradiance	taub		0.466	0.575	0.691	0.654	0.504	0.458	0.446	0.464	0.507	0.542	0.455
			taud	2.113	1.839	1.619	1.740	2.169	2.326	2.356	2.286	2.151	2.010	2.230
		Ebn at Noon		812	740	672	702	806	836	847	838	803	758	818
			Edh at Noon	150	207	266	237	153	129	125	135	154	172	133
		RadAvg		4.64	5.07	5.14	5.60	5.26	4.83	4.56	4.49	4.63	4.86	4.84
(45)	All-Sky Solar Radiation	RadStd		0.32	0.28	0.29	0.28	0.37	0.37	0.32	0.32	0.26	0.31	0.19

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page